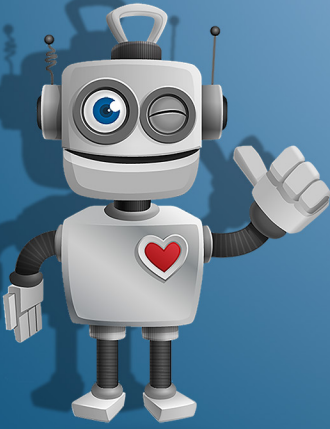


تقدم لكم اللجنة الاكاديمية في فريق

Mechatronics spectrum

امتحان سابق لمادة

اجهزة الحماية و التدكم الكهربائية



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0785060940

Electrical Protection & Control Devices. First Examination

Summer Semester, 2015

28/06/2015

Time: 50 minutes

Name: _____

28
20

Q1: What is the meaning of the commonly used abbreviations (5 points)

BR	Break relay ✓
OL	over load ✓
PB	push button ✓
CR	Control relay ✓
DISC	Disconnected switch ✓
DPDT	double pole double throw ✓
FTS	Foot switch ✓
GRD	Ground ✓
INTLK	inter lock ✓
JB	Junction Box ✓
UV	under voltage ✓
VFD	variable frequency Drive ✓

Q2: Specify whether each of the following statements are true or false by writing either "TRUE" or "FALSE" in the blank answer space preceding each statement. (8 points)

Note – answer spaces containing only a "T" or an "F" will be graded as "incorrect".

1. A Normally Open (NO) Pushbutton contains one or more sets of contacts that "close" while the button is pressed and then return back to an "open" state when the button is released. True
2. A Normally Closed (NC) Auxiliary Contact is often used to "hold-in" a contactor after the "start" pushbutton in a motor control system is pressed and then released. False
3. A contactor's Auxiliary Contacts function in the exact same manner as its Main Contacts, although the auxiliary contacts typically have a much lower current rating than that of the main contacts. True
4. An Overload Relay is designed to protect a motor from an overload due to a high supply voltage magnitude while the machine is driving up to its rated mechanical load. True



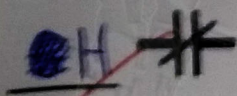
Although an Overload Relay uses an Off-Delay Timer to delay shutting-down a motor during start-up", it is able to bypass the timer and immediately shut down the motor if it detects an overload condition while the motor is operating near rated speed. ~~True~~ **False**

6. Although contactors are designed to be able to disconnect an energized motor from its supply lines (i.e. - shut-down the motor) even during overload conditions, contactors are not designed to interrupt short-circuit or other similarly large currents. ~~True~~ **False**

7. While a motor is driving its rated load, a moderate increase in the supply voltage magnitude will likely result in an overload condition where as a moderate decrease in the supply voltage magnitude may slow-down the motor but will not result in an overload. ~~False~~ **True**

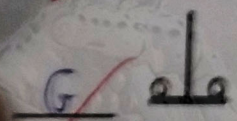
8. Starting a large induction motor by applying full rated voltage to its terminals will result in large inrush currents and an associated developed torque surge during start-up, with both the line currents and the developed torque typically reaching magnitudes that are four to ten times larger than their rated value. ~~False~~ **True**

Q3: Match the symbols shown in the left-hand column with the devices listed in the right-hand column by writing the letter associated with the correct device next to the appropriate symbol. (4 points)

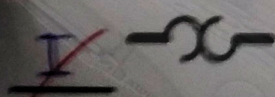


A) Main Contactor's Field Coil

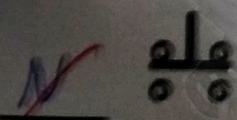
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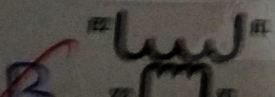
B) Transformer



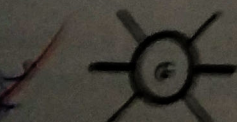
C) Circuit Breaker



D) Two Position Switch



E) Control Relay's Field Coil



F) Indicator Lamp



G) Normally Closed Pushbutton



H) Normally Closed Contact



I) Overload Relay's Heater



J) Fuse

K) Pressure Switch

L) Normally Open Pushbutton

M) Normally Open Contact

N) None of the above

